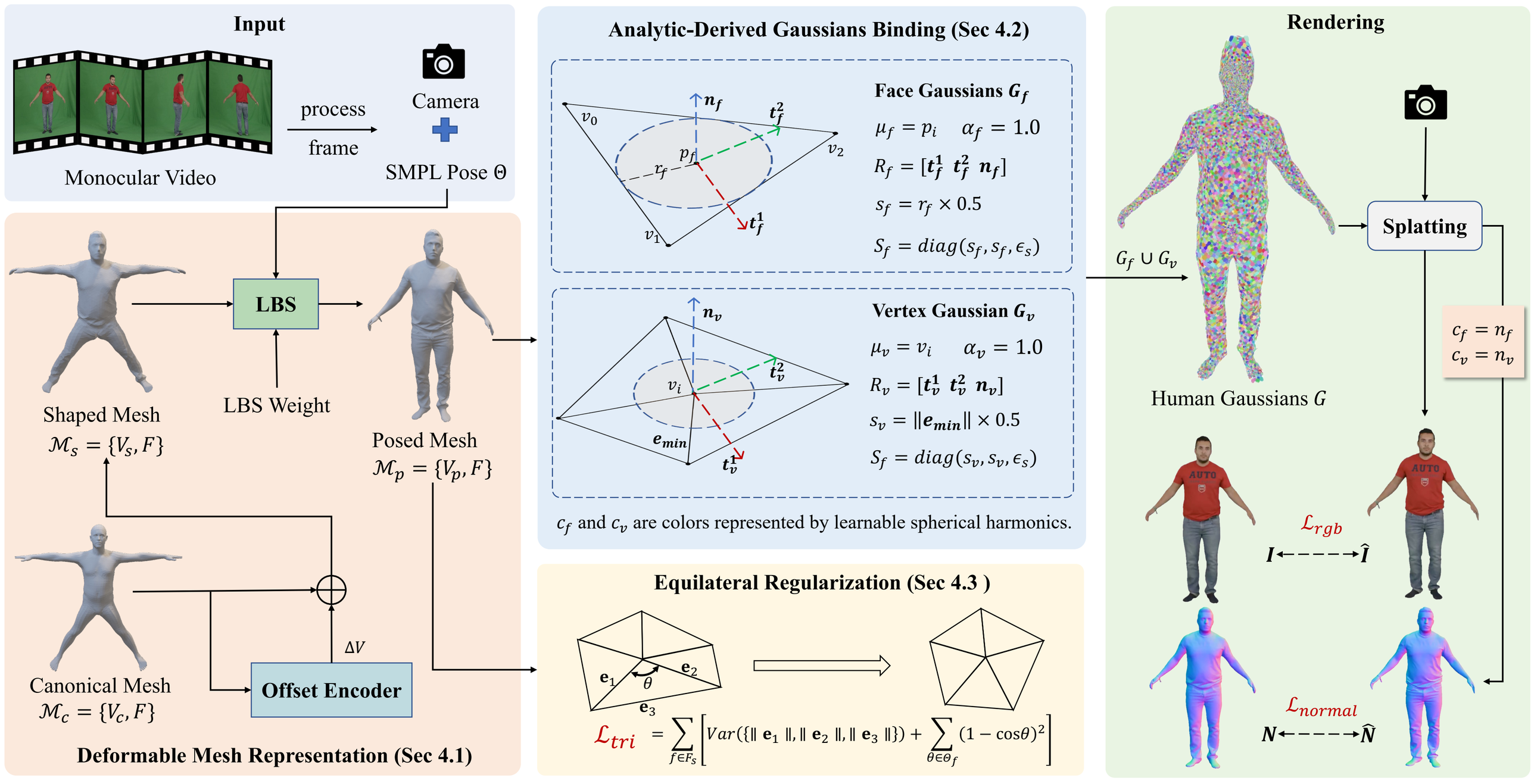
Input
Monocular Video
process
frame
Camera
SMPL Pose Θ
LBS
LBS Weight
Shaped Mesh
$\mathcal{M}_s = \{V_s, F\}$
Posed Mesh
$\mathcal{M}_p = \{V_p, F\}$
Canonical Mesh
$\mathcal{M}_c = \{V_c, F\}$
ΔV
Offset Encoder
Deformable Mesh Representation (Sec 4.1)
Analytic-Derived Gaussians Binding (Sec 4.2)
Face Gaussians G_f
$\mu_f = p_i \quad \alpha_f = 1.0$
$R_f = [\boldsymbol{t}_f^1 \ \boldsymbol{t}_f^2 \ \boldsymbol{n}_f]$
$s_f = r_f \times 0.5$
$S_f = diag(s_f, s_f, \epsilon_s)$
Vertex Gaussian G_v
$\mu_v = v_i \quad \alpha_v = 1.0$
$R_v = [\boldsymbol{t}_v^1 \ \boldsymbol{t}_v^2 \ \boldsymbol{n}_v]$
$s_v = \|\boldsymbol{e}_{min}\| \times 0.5$
$S_f = diag(s_v, s_v, \epsilon_s)$
c_f and c_v are colors represented by learnable spherical harmonics.
Equilateral Regularization (Sec 4.3)
$\mathcal{L}_{tri} = \sum_{f \in F_s} \left[Var(\{\| \mathbf{e}_1 \|, \| \mathbf{e}_2 \|, \| \mathbf{e}_3 \|\}) + \sum_{\theta \in \Theta_f} (1 - \cos\theta)^2 \right]$
Rendering
$G_f \cup G_v$
Human Gaussians G
Splatting
$c_f = n_f$
$c_v = n_v$
$\mathcal{L}_{rgb}$
$I \longleftrightarrow \hat{I}$
$\mathcal{L}_{normal}$
$N \longleftrightarrow \hat{N}$